

Lecture 5 – Abstract Interpretation

SWS128: Software Analysis

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2026 Fall

- A **dataflow analysis**: data and a join, a transfer per edge, a direction, an initial value. One engine, `solve`, runs them all.
- We said its answer is **sound** and the **least** one. We never said why.
- Today: the theory behind those two words. Dataflow analysis turns out to be one instance of it.

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Abstract Domains

2. Two Analyses

Step by Step

3. Galois Connections

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Running It

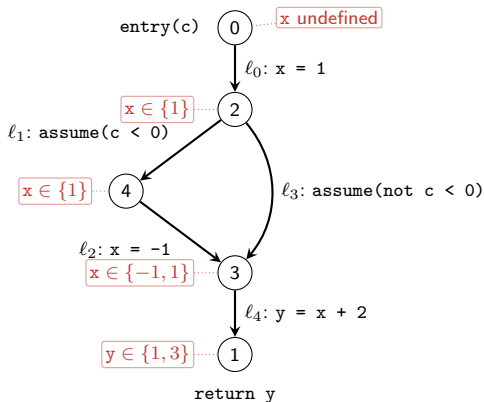
```
def f(c):  
    x = 1  
    if c < 0:  
        x = -1  
    y = x + 2  
    return y
```

One run per input:

c	returns
-3	1
0	3
5	3
$\vdots$	$\vdots$

The interpreter answers for **one** c . An analysis must answer for **every** c at once: *is y always positive?*

At each point, the values x and y can have, over **all** runs:



The exact answer: y is always positive. The equations are the dataflow ones, $C(q) = \bigcup_{p \rightarrow q} f(C(p))$, over sets of concrete values.

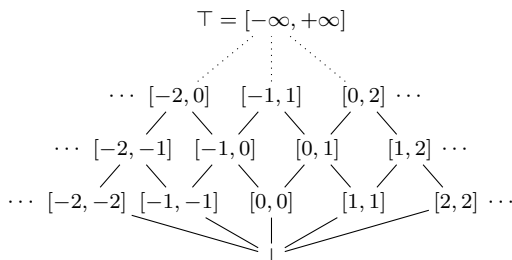
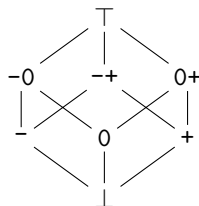
```
def g(n):  
    x = 0  
    while x < n:  
        x = x + 1  
    return x
```

The values of x at the loop head, one iteration of the equations at a time:

$$\{0\} \subset \{0, 1\} \subset \{0, 1, 2\} \subset \dots$$

- The chain never stabilizes: n can be any number, so the exact set is **infinite**, and no finite number of iterations reaches it.
- The collecting semantics is exact but, in general, **cannot be computed**. We need a **finite** description of each set.

Replace each set of values by a **finite description** of it. We draw each domain as a **Hasse diagram**: a line goes up from smaller to larger.



Sign: which of $-$, 0 , $+$ can occur

Interval: $[l, u]$, ordered by inclusion

Set of values	As a sign	As an interval
$\{1\}$	$+$	$[1, 1]$
$\{-1, 1\}$	$-+$	$[-1, 1]$
$\{0, 1, 2, \dots\}$	$0+$	$[0, +\infty]$

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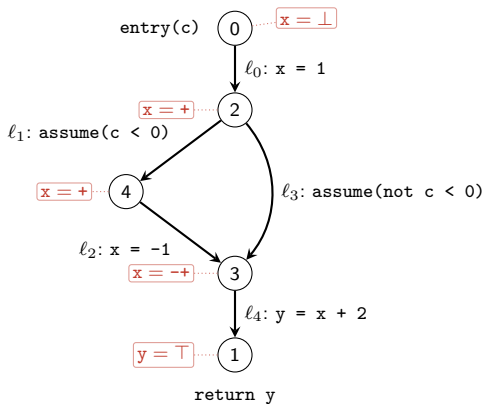
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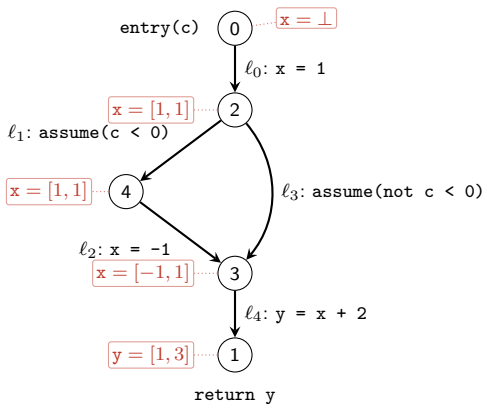
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ℓ_4 : adding + to $-+$ gives $y = \top$, since $-1 + 2 > 0$ but $-5 + 2 < 0$. *Is y positive? Sign **cannot** tell.*



ℓ_4 : adding $[2, 2]$ to $[-1, 1]$ gives $y = [1, 3]$. *Is y positive?* Interval says **yes**.

	concrete	sign	interval
x at 3	$\{-1, 1\}$	$-+$	$[-1, 1]$
y at 1	$\{1, 3\}$	$\top$	$[1, 3]$
y positive?	yes	<i>don't know</i>	yes

- Both are **correct**: every real value is inside what they say.
- Sign kept “not zero” but not *how big*: a negative plus 2 can be anything. Interval kept $[-1, 1]$, enough to add 2 and stay positive.
- Choosing a domain is choosing what you can prove.

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Two functions connect sets of values and the abstract domain:

- $\gamma(a)$: the set of values a describes.

$$\gamma(+) = \{1, 2, 3, \dots\} \quad \gamma([1, 3]) = \{1, 2, 3\}$$

- $\alpha(S)$: the best description of S .

$$\alpha(\{-1, 1\}) = -+ \text{ (sign)} \quad \alpha(\{-1, 1\}) = [-1, 1] \text{ (interval)}$$

They form a **Galois connection** when

$$\alpha(S) \sqsubseteq a \iff S \subseteq \gamma(a)$$

that is, a describes S correctly exactly when a is above $\alpha(S)$.

$$S = \{1, 100\} \quad \alpha(S) = [1, 100] \quad \gamma(\alpha(S)) = \{1, 2, \dots, 100\}$$

- Intervals cannot say “1 or 100”. They say “between 1 and 100”, and 98 values that never occur come with it.
- Always $S \subseteq \gamma(\alpha(S))$: abstraction adds values, never removes them. That is the price of computability.

Each operation needs an abstract version. For sign, define it on the three signs:

$+^\sharp$	-	0	+
-	-	-	$\top$
0	-	0	+
+	$\top$	+	+

$\times^\sharp$	-	0	+
-	+	0	-
0	0	0	0
+	-	0	+

- Any other element is a set of signs: take every pair, then join.

$$- + +^\sharp + = (- +^\sharp +) \sqcup (+ +^\sharp +) = \top \sqcup + = \top$$

- Sound:** $f(\gamma(a)) \subseteq \gamma(f^\sharp(a))$. A negative times a negative is always positive, so $- \times^\sharp - = +$.
- Interval: $[a, b] +^\sharp [c, d] = [a + c, b + d]$.

The analysis is **sound** when, at every point,

$$C(n) \subseteq \gamma(A(n))$$

	C	$\gamma(\text{sign})$	$\gamma(\text{interval})$
x at 2	$\{1\}$	$\{1, 2, \dots\} \checkmark$	$\{1\} \checkmark$
x at 3	$\{-1, 1\}$	$\mathbb{Z} \setminus \{0\} \checkmark$	$\{-1, 0, 1\} \checkmark$
y at 1	$\{1, 3\}$	$\mathbb{Z} \checkmark$	$\{1, 2, 3\} \checkmark$

- Each transfer is sound and each join is sound, so the result is.
- This is the contract of every analysis in this course.

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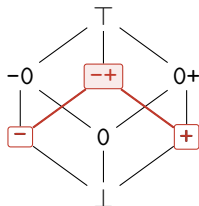
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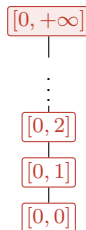
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Pick any elements. Is there a **lowest** element above all of them?



Lowest above $-$ and $+$: $-+$

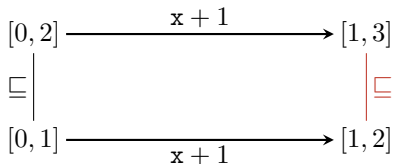


Lowest above $[0, n]$ for all n : $[0, +\infty]$

- A **complete lattice**: the answer is always yes. The lowest element above a set is its **join** $\sqcup$.

F is **monotone** if a larger input never gives a smaller output:

$$a \sqsubseteq b \implies F(a) \sqsubseteq F(b)$$



- Every transformer we built is monotone: $+\sharp$, $\times\sharp$, and $\sqcup$. So is F , which only combines them.

All the equations together are one function F on the table of abstract values. The answer is a table A with $F(A) = A$.

Theorem (Knaster–Tarski)

If $\mathbb{D}$ is a complete lattice and F is monotone, then F has a **least fixpoint**, $\text{lfp } F$.

- Every fixpoint is sound. The least one is the most precise.
- So the answer we want exists, and there is exactly one.

Start from $\perp$ and apply F until nothing changes:

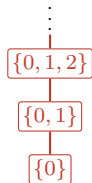
$$\perp \sqsubseteq F(\perp) \sqsubseteq F^2(\perp) \sqsubseteq \dots \quad \text{lf}_p F = \bigsqcup_i F^i(\perp)$$

x at the loop head of g , in sign:

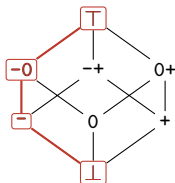
	F^0	F^1	F^2	F^3
x	$\perp$	0	$0 \sqcup (0 +^\# +) = 0+$	$0 \sqcup (0+ +^\# +) = 0+$

- Two steps and it is stable at $0+$. The concrete chain $\{0\}, \{0, 1\}, \dots$ never was.
- The worklist computes this same chain: same fixpoint, less work.

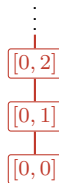
The **height** of a domain is the length of its longest chain $a_0 \sqsubseteq a_1 \sqsubseteq \dots$.



Concrete: infinite



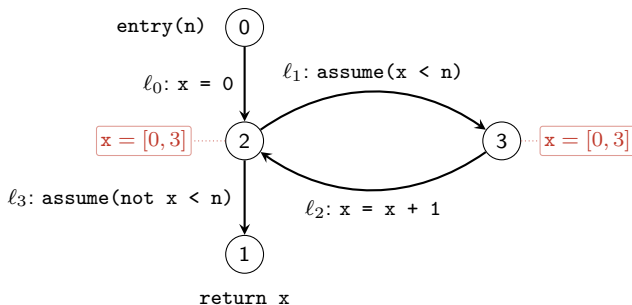
Sign: 3



Interval: infinite

- Each Kleene step either stays or moves up. With finite height, it must stop. Sign is guaranteed to terminate.
- Concrete and Interval have an infinite chain: **no guarantee**.

The same loop g , in interval:



Each lap adds one: $[0, 0] \subseteq [0, 1] \subseteq [0, 2] \subseteq \dots$ **never stops.**

- Sign stopped at $0+$ after two steps. Interval has **infinite height**: the chain has no end.
- The engine gives up (RuntimeError: no fixpoint). The fix, **widening**, is the next lecture.

Abstract interpretation	Dataflow analysis
abstract domain $\mathbb{D}$	sets of facts, ordered by $\subseteq$
abstract transformer $f^\sharp$	$\text{gen} \cup (X \setminus \text{kill})$
join $\sqcup$	$\cup$ or $\cap$
$\text{lfp } F$	what <code>solve</code> returned
soundness $C \subseteq \gamma(A)$	assumed, never stated

- A recipe for a new analysis: pick $\mathbb{D}$, define γ , write a sound $f^\sharp$ per operation, run the engine.

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```
class Domain:
    bot: object
    top: object

    def le(self, a, b) -> bool: raise NotImplementedError
    def join(self, a, b): raise NotImplementedError
    def meet(self, a, b): raise NotImplementedError
    def alpha(self, ns: set[int]): raise NotImplementedError
    def gamma(self, a) -> str: raise NotImplementedError
    def of_num(self, n: int): raise NotImplementedError
    def binop(self, op: str, a, b): raise NotImplementedError
    def unop(self, op: str, a): raise NotImplementedError

    def widen(self, a, b): return self.join(a, b)
    def narrow(self, a, b): return self.meet(a, b)
```

- The lattice (`le`, `join`, `meet`), the Galois connection (`alpha`, `gamma`), and the transformers (`of_num`, `binop`, `unop`).
- `widen` and `narrow` are for the next lecture.

```
# an element is a set of the three signs; its name is the key
ATOMS = {BOT: "", NEG: "-", ZERO: "0", POS: "+",
         NONPOS: "-0", NONNEG: "0+", NONZERO: "-+", TOP: "-0+"}
NAME = {frozenset(v): k for k, v in ATOMS.items()}

ADD = {("-", "-"): "-", ("-", "0"): "-", ("-", "+"): "-0+",
       ("0", "0"): "0", ("0", "+"): "+", ("+", "+"): "+"}

class Sign(Domain):
    def join(self, a, b):
        return NAME[frozenset(ATOMS[a] + ATOMS[b])]
    def binop(self, op, a, b):
        ...
        out = ""
        for x in ATOMS[a]:
            for y in ATOMS[b]:
                out += table.get((x, y)) or table[(y, x)]
        return NAME[frozenset(out)]
```

- Join is union of signs. An operation is the three-sign table, applied to every pair.

```
>>> from syntax import parse
>>> from cfg import build
>>> from domain import Sign, analyze, returned
>>> src = """
... def f(c):
...     x = 1
...     if c < 0:
...         x = -1
...     y = x + 2
...     return y
... """
>>> g = build(parse(src))
>>> d = Sign()
>>> d.join("-", "+"), d.binop("+", "-+", "+")
('+-+', 'top')
>>> res = analyze(g, d)
>>> res[3]["x"], returned(g, d, res)
('+-+', 'top')
```

- The same answers as the step-by-step: x is $-+$ at the merge, y is $\top$.
The interval domain is the next homework.

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- Widening and Narrowing

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